ConfigMap from Literals

1. **kubectl create configmap**
   Create a ConfigMap via CLI
   `kubectl create configmap app-config \`

2. **--from-literal**
   Pass key=value pairs inline
   `--from-literal=APP_ENV=production`

3. **--from-literal**
   Multiple keys supported
   `--from-literal=APP_PORT=8080`

4. **kubectl describe configmap**
   Stored as plain text — no encoding
   `kubectl describe configmap app-config`

🔲 ConfigMap from a File

1. **nginx config file**
   Write a config with /health endpoint
   `location /health { return 200 "healthy"; }`

2. **kubectl create configmap**
   Create ConfigMap from the file
   `kubectl create configmap nginx-config \`

3. **--from-file**
   Key = filename, value = file contents
   `--from-file=default.conf=default.conf`

4. **kubectl get configmap -oyaml**
   Verify the full file is stored
   `kubectl get configmap nginx-config -o yaml`

ConfigMaps in Pods

1. envFrom + configMapRef

Injects ALL keys as env variables

```
envFrom:
- configMapRef:
```

2. volumeMounts

Mounts config file at path in container

```
mountPath: /etc/nginx/conf.d
```

3. kubectl exec -- curl

Test that mounted nginx config works

```
kubectl exec <pod> -- curl -s http://localhost/health
```

4. envFrom vs Volume

Env = frozen at start. Volume = live updates.

Volume mounts update ~60s. Env vars never.

Create a Secret

1. **kubectl create secret**

   Create a generic Secret object

   ```
   kubectl create secret generic db-credentials \
   ```

2. **--from-literal**

   Store sensitive key-value pairs

   ```
   --from-literal=DB_PASSWORD=s3cureP@ssw0rd
   ```

3. **kubectl get secret -o yaml**

   Values appear base64-encoded in output

   ```
   kubectl get secret db-credentials -o yaml
   ```

4. **echo | base64 --decode**

   base64 is encoding — NOT encryption

   ```
   echo "<base64-value>" | base64 --decode
   ```

Secrets in Pods

1. **secretKeyRef**
   Inject one Secret key as env variable
   ```
   valueFrom:
     secretKeyRef:
   ```

2. **Secret volume mount**
   Mount entire Secret as files — read only
   ```
   mountPath: /etc/db-credentials
   readOnly: true
   ```

3. **cat mounted file**
   Each key is a file with plaintext value
   ```
   cat /etc/db-credentials/DB_PASSWORD
   ```

4. **Real security**
   RBAC + tmpfs + encryption at rest in etcd
   base64 alone protects nothing. Use RBAC.

☐ Live ConfigMap Updates

1. **kubectl create configmap**
   Create ConfigMap with initial value
   kubectl create configmap live-config \

2. **Pod reads file in loop**
   Container reads the mounted file every 5s
   while true; do cat /config/message; sleep 5; done

3. **kubectl patch configmap**
   Update value without touching the Pod
   kubectl patch configmap live-config --type merge \

4. **Auto-propagation ~60s**
   File changes live. No restart needed.
   Env vars stay frozen. Volume mounts go live.